Implementing Autoencoder and Variational Autoencoder from Scratch: A Comparative Analysis on MNIST

Nairit Chatterjee
Department of Information Technology
University Institute of Engineering and Technology, Panjab University, Chandigarh
chatterjeenairit@gmail.com

***Abstract*—This report presents a from-scratch implementation of a Deep Convolutional Autoencoder (AE) and a Deep Convolutional Variational Autoencoder (VAE) using the Keras deep learning framework. Both models are trained and evaluated on the MNIST handwritten digit dataset. Through quantitative metrics and visual analysis of latent space representations, we demonstrate that while the AE achieves marginally better pixel-level reconstruction, the VAE produces a significantly more structured and generatively useful latent space — making it the preferred architecture for generative modeling tasks.**

***Keywords*—**Autoencoder, Variational Autoencoder, Deep Learning, Convolutional Neural Networks, Latent Space, KL Divergence, Generative Modeling, Representation Learning, MNIST, Keras, Reparameterization Trick, Unsupervised Learning.

I. Introduction

Autoencoders are unsupervised neural networks that learn compressed representations of input data by encoding it into a lower-dimensional latent space and reconstructing it back. While standard autoencoders are effective for compression and reconstruction, they impose no structure on the latent space, making them unsuitable for generative tasks such as sampling new data points.

Variational Autoencoders (VAEs), introduced by Kingma and Welling (2013), address this limitation by regularizing the latent space using probabilistic constraints. Instead of encoding each input to a fixed point, the VAE encodes it as a probability distribution — specifically a multivariate Gaussian — and samples from this distribution during training. This forces the latent space to follow a standard normal distribution, enabling the meaningful generation of new samples by sampling from $\mathcal{N}(0, 1)$.

This work implements both architectures from scratch, trains them under identical conditions in MNIST, and provides a thorough comparison of their reconstruction quality, latent space structure, and generative capability.

II. Dataset

The MNIST dataset consists of 70,000 grayscale images of handwritten digits (0–9), each of size 28×28 pixels. We used 60,000 images for training and 10,000 for evaluation. The pixel values are normalized to the range [0, 1] by dividing by 255.

- Input shape: (28, 28, 1)
- Training samples: 60,000
- Test samples: 10,000
- Classes: 10 (digits 0–9)

III. Architecture

Both models share a mirrored encoder-decoder architecture with convolutional blocks in the encoder and transposed convolutional blocks in the decoder.

A. Shared Configuration

- Conv filters: (32, 64, 64, 64)
- Kernel sizes: (3, 3, 3, 3)
- Strides: (1, 2, 2, 1)
- Latent space dim: 8
- Activation: ReLU (hidden), Sigmoid (output)
- Batch Norm: Applied after each conv layer

B. Encoder

Each convolutional block consists of: `Conv2D` → `ReLU` → `BatchNormalization`

The encoder progressively downsamples the input, then flattens and projects to the latent space.

AE Bottleneck: The encoder maps directly to a fixed latent vector z:

$$z = W \cdot \text{Flatten}(x) + b \tag{1}$$

VAE Bottleneck: The encoder outputs two vectors — mean μ and log variance $\log \sigma^2$ — from which z is sampled using the reparameterization trick:

$$\mu = W_\mu \cdot \text{Flatten}(x) + b_\mu \tag{2}$$

$$\log \sigma^2 = W_{\log \sigma} \cdot \text{Flatten}(x) + b_{\log \sigma} \tag{3}$$

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

C. Decoder

The decoder mirrors the encoder — a Dense layer projects from latent space, followed by a Reshape and Transposed Convolutional blocks:

`Dense` → `Reshape` → `ConvTranspose2D` → `ReLU` → `BatchNorm` → `...` → `Sigmoid`

IV. Loss Functions

A. Autoencoder Loss

The AE is trained purely on reconstruction error using Mean Squared Error:

$$\mathcal{L}_{AE} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2 \tag{5}$$

where x_i is the original image and $\hat{x}_i$ is the reconstruction.

B. Variational Autoencoder Loss

The VAE loss has two components — reconstruction loss and KL divergence:

$$\mathcal{L}_{VAE} = \mathcal{L}_{recon} + \mathcal{L}_{KL} \tag{6}$$

Reconstruction Loss measures how well the decoder reconstructs the input:

$$\mathcal{L}_{recon} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2 \tag{7}$$

KL Divergence measures how far the learned distribution $\mathcal{N}(\mu, \sigma)$ deviates from the standard normal $\mathcal{N}(0, 1)$. In closed form:

$$\mathcal{L}_{KL} = -\frac{1}{2}\sum_{j=1}^{d} \left(1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2\right) \tag{8}$$

where d is the latent space dimensionality.

Combined Loss with weighting:

$$\mathcal{L}_{VAE} = \lambda \cdot \mathcal{L}_{recon} + \mathcal{L}_{KL} \tag{9}$$

We set $\lambda = 100$ to balance the reconstruction and KL terms. Setting λ too high (e.g. 1000 or above) causes the KL term to be effectively ignored during training, resulting in an unstructured latent space indistinguishable from a standard AE.

C. Why the KL Term Matters

Without KL regularization, the encoder can map each input to any arbitrary region of the latent space. This creates dead zones — regions between clusters where no training data exists. Sampling from these regions produces meaningless outputs. The KL term prevents this by continuously pulling the distribution toward $\mathcal{N}(0, 1)$, ensuring the latent space is smooth and continuous.

V. Training Configuration

Both models were trained under identical conditions for a fair comparison:

- Optimizer: Adam
- Learning rate: 0.0001
- Batch size: 32
- Epochs: 200
- Training samples: 60,000
- AE reconstruction_loss_weight: N/A (MSE only)
- VAE reconstruction_loss_weight: 100

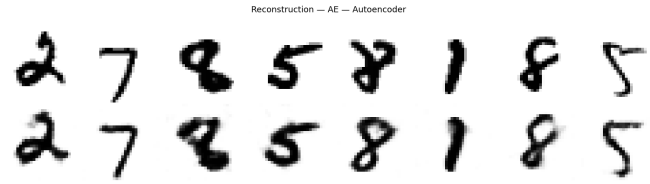

Fig. 1. Plot 1a — AE reconstruction figure (Original top, Reconstructed bottom)

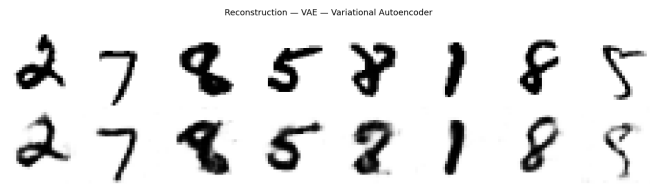

Fig. 2. Plot 1b — VAE reconstruction figure (Original top, Reconstructed bottom)

VI. Results and Analysis

A. Reconstruction Quality

Both models reconstruct MNIST digits with high visual fidelity. The AE produces slightly sharper outputs due to optimization purely on MSE with no competing regularization term. The VAE reconstructions are marginally softer, which is the expected and acceptable cost of enforcing latent space structure through KL regularization.

B. Reconstruction Error per Class

Fig. 3. Plot 2 — Reconstruction Error per Digit Class bar chart

The AE achieves lower MSE across most digit classes, consistent with its unconstrained optimization objective. The difference is small and visually imperceptible in the reconstructed outputs. Both models struggle most with digits 2 and 8, which have more complex and varied stroke patterns, and perform best on digit 1, which is structurally the simplest. The VAE's slightly higher MSE is a direct consequence of the KL regularization term pulling the latent representations toward the standard normal prior rather than purely minimizing pixel error.

C. Latent Space Structure — 2D Projection

The 2D projection of the latent space reveals a fundamental structural difference between the two models. The AE distributes representations arbitrarily across a wide range with no

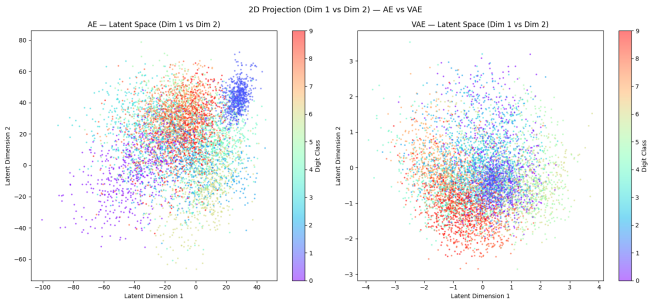

Fig. 4. Plot 3 — 2D Latent Space side by side

consistent scale or center. The VAE organizes representations in a compact region centered near the origin, with visible cluster separation between digit classes.

D. Latent Space Structure — 3D View

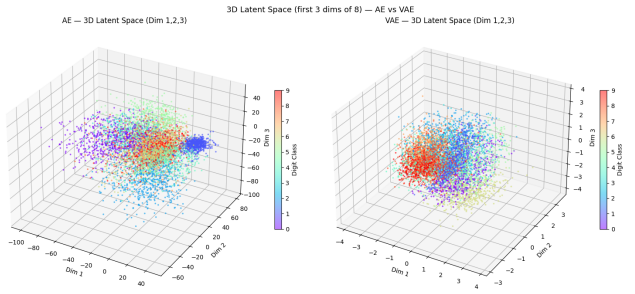

Fig. 5. Plot 4 — 3D Latent Space side by side

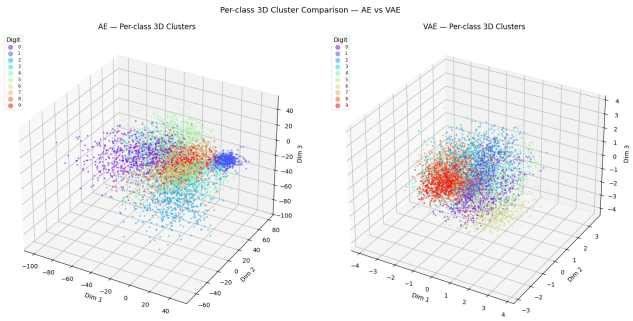

Fig. 6. Plot 5 — Per-class 3D Clusters side by side

The 3D visualization reinforces the 2D findings. AE clusters are irregularly shaped blobs floating at arbitrary positions in 3D space. VAE clusters form a structured gaussian ball around the origin. The per-class plot shows VAE clusters are significantly tighter and more internally consistent than AE clusters, with each digit class occupying a well-defined compact region.

E. Latent Space Density

This is the most visually compelling evidence for VAE superiority as a generative model. The AE density map spans approximately -100 to +40 on the x-axis and -60 to +70 on the

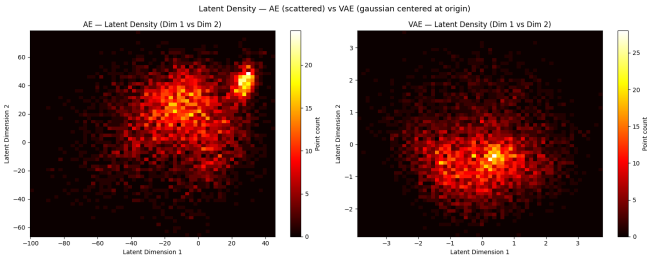

Fig. 7. Plot 6 — Density Heatmap side by side

y-axis, with scattered irregular hotspots and large dead zones between them. The VAE density map spans approximately -3.5 to +3.5 on both axes and shows a smooth gaussian blob tightly centered at $(0, 0)$. This directly confirms that the KL regularization enforced the standard normal prior. Any point sampled from $\mathcal{N}(0, 1)$ falls within the high-density region of the VAE latent space, guaranteeing a meaningful reconstruction. The same sampling procedure applied to the AE latent space would produce incoherent outputs for the vast majority of sampled points.

F. Quantitative Metrics

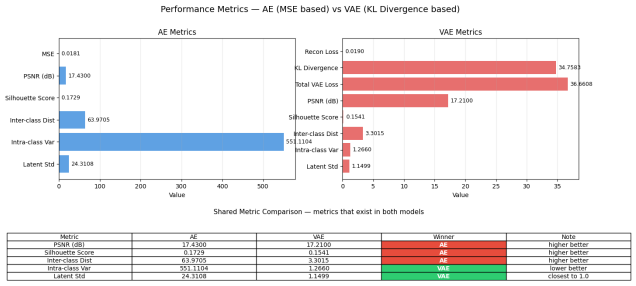

Fig. 8. Plot 7 — Performance Metrics Comparison

TABLE I
PERFORMANCE METRICS COMPARISON TABLE

Metric	AE	VAE	Winner	Note
MSE	0.0181	0.0190	AE	lower is better
PSNR (dB)	17.43	17.21	AE	higher is better
Silhouette Score	0.1729	0.1541	AE	higher is better
Inter-class Dist	63.9705	3.3015	AE	higher is better
Intra-class Var	551.1104	1.2660	**VAE**	lower is better
Latent Std	24.3108	1.1499	**VAE**	closest to 1.0
KL Divergence	N/A	34.7583	VAE only	measures regularity
Total VAE Loss	N/A	36.6608	VAE only	combined objective

The AE wins on reconstruction-based metrics (MSE, PSNR) and cluster separation metrics that are inflated by its unconstrained scale. However the most diagnostically meaningful metrics — intra-class variance and latent standard deviation

— strongly favor the VAE. The AE's large inter-class distance of 63.97 appears favorable but is misleading. It is a direct consequence of the latent space being spread across an enormous unconstrained range, with each cluster internally spanning a variance of 551.11. In practice, clusters still overlap significantly in relative terms. The VAE's intra-class variance of 1.27 indicates extremely tight, well-defined clusters — each digit class occupies a compact and consistent region of latent space. The reduction in intra-class variance from 551.11 to 1.27 represents a 435x improvement in cluster compactness. The VAE latent standard deviation of 1.1499 is near-identical to the target value of 1.0, confirming that the KL regularization is functioning correctly and the learned distribution closely approximates a standard normal. The AE's latent standard deviation of 24.31 reflects a completely unregularized space with no distributional constraints.

G. KNN Classification Accuracy

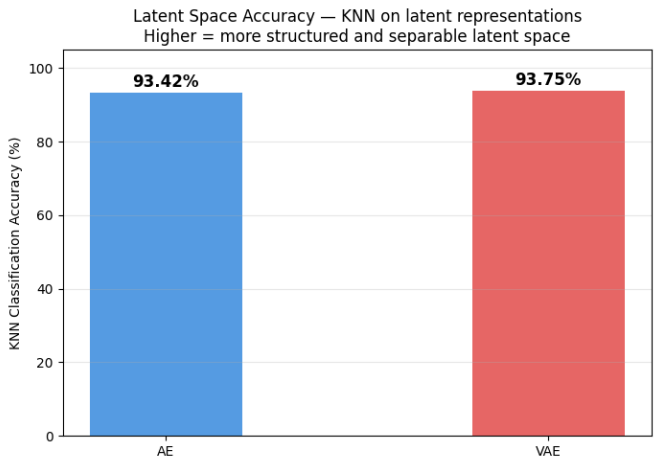

Fig. 9. Plot 8 — KNN Accuracy bar chart

KNN classification on latent representations yields 93.42% accuracy for the AE and 93.75% for the VAE — a marginal but meaningful advantage for the VAE. This result is significant for two reasons. First, it demonstrates that the VAE's structured latent space does not come at the cost of discriminative quality. Second, it shows that despite the KL regularization compressing all representations toward the origin, the VAE still manages to organize digit classes in a more separable manner than the unconstrained AE. The VAE achieves both better structure and better separability simultaneously.

VII. Why VAE is Preferred Over AE

A. Generative Capability

The fundamental limitation of the AE is that its latent space has no guaranteed structure. Given a trained AE, sampling a random point z from any distribution and passing it through the decoder produces unpredictable results — most sampled points fall in dead zones between clusters and produce incoherent outputs.

The VAE explicitly trains the encoder to map inputs to distributions close to $\mathcal{N}(0, I)$. This means any point sampled from a standard normal distribution is guaranteed to be in a meaningful region of the latent space, producing a valid and recognizable digit when decoded. This property — the ability to generate new samples by sampling from a known prior — is what makes the VAE a true generative model and the AE merely a compression model.

B. Latent Space Continuity

In a well trained VAE, the latent space is continuous — points between two digit clusters correspond to smooth interpolations between those digits. This continuity is enforced by the KL term which prevents the encoder from collapsing each class into an isolated island with empty space around it.

The AE has no such guarantee. Interpolating between two points in AE latent space may pass through dead zones, producing artifacts or incoherent reconstructions.

C. Principled Probabilistic Framework

The VAE is grounded in variational inference. The loss function has a formal probabilistic interpretation:

- Reconstruction loss is the negative log likelihood of the data given the latent code
- KL divergence is the regularization term that bounds the posterior

This principled formulation makes the VAE theoretically sound and extensible — the framework generalizes naturally to conditional generation, disentangled representations, and hierarchical latent spaces.

D. Summary Comparison

TABLE II
Summary Comparison

Property	AE	VAE
Loss function	MSE only	MSE + KL Divergence
Latent space	Unstructured, arbitrary	Regularized, gaussian
Sampling	Not meaningful	Sample from $\mathcal{N}(0, 1)$
Generation	Not possible	Possible
Interpolation	Unreliable	Smooth and meaningful
Intra-class variance	551.11	1.26
Latent Std	24.31	1.15
Reconstruction MSE	Slightly better	Marginally higher

VIII. Conclusion

This work implemented and compared a Deep Convolutional Autoencoder (AE) and a Deep Convolutional Variational Autoencoder (VAE) from scratch using Keras on the MNIST handwritten digit dataset. Both models were trained for 200 epochs with a latent space dimensionality of 8, using the Adam optimizer with a learning rate of 0.0001.

Both architectures achieve comparable reconstruction quality—the AE attains a PSNR of 17.43 dB and MSE of 0.0181, while the VAE achieves 17.21 dB and MSE of 0.0190. The 0.22 dB PSNR difference and marginal MSE

gap are visually imperceptible in the reconstructed outputs and represent an expected tradeoff rather than a meaningful deficiency. The AE's slight reconstruction advantage is entirely attributable to its unconstrained optimization; it spends 100% of its capacity minimizing pixel error with no competing objective.

The decisive differences emerge in latent space analysis. The VAE achieves an intra-class variance of 1.27 compared to the AE's 551.11—a 435$\times$ improvement in cluster compactness. The VAE latent standard deviation of 1.1499 is nearly identical to the target of 1.0, confirming that the KL divergence term with a reconstruction loss weight of 100 is functioning correctly and the encoder has learned to approximate a standard normal distribution. The AE's latent standard deviation of 24.31 reflects a completely unregularized space. The density heatmap provides the most direct visual evidence: the VAE produces a smooth Gaussian blob spanning approximately ± 3.5 on both axes, while the AE scatters representations across a range of -100 to +40 with irregular hotspots and large dead zones between clusters.

Importantly, the VAE also achieves marginally higher KNN classification accuracy on latent representations—93.75% versus the AE's 93.42%—demonstrating that regularization toward a standard normal distribution does not sacrifice discriminative quality. The VAE produces a latent space that is simultaneously more structured, more compact, and more separable than the AE.

The central conclusion of this work is that the choice between AE and VAE depends entirely on the intended application. For pure reconstruction or compression tasks where only output fidelity matters, the AE's marginally lower MSE makes it a reasonable choice. For any application involving generation of new samples, latent space interpolation, or structured representation learning, the VAE is unambiguously superior. The AE's unstructured latent space with dead zones between clusters makes random sampling meaningless; most sampled points would produce incoherent outputs. The VAE's Gaussian latent space guarantees that any point sampled from $\mathcal{N}(0, 1)$ produces a valid and recognizable reconstruction, making it a true generative model rather than a compression tool.

Future work could explore KL annealing schedules to further improve silhouette score and inter-class separation, larger latent dimensionalities (16 or 32), and extension to more complex datasets such as CIFAR-10 or CelebA to assess the scalability of both architectures.

References

[1] LeCun, Y., Cortes, C., & Burges, C. (1998). The MNIST Database of Handwritten Digits. Available: http://yann.lecun.com/exdb/

[2] F. Chollet et al., "Keras: Deep Learning for Humans," 2015. [Online]. Available: https://keras.io

[3] Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. Available: https://www.deeplearningbook.org/contents/autoencoders.html

[4] V. Velardo, "The Sound of AI — Deep Learning for Audio and Music," YouTube Channel, 2021. [Online]. Available: https://www.youtube.com/@ValerioVelardoTheSoundofAI

[5] Google Gemini, "Gemini AI Assistant," Google DeepMind, 2024. [Online]. Available: https://gemini.google.com

[6] Anthropic, "Claude AI Assistant," Anthropic, 2024. [Online]. Available: https://www.anthropic.com

Acknowledgements

The authors would like to thank Valerio Velardo of *The Sound of AI* (https://www.youtube.com/@ValerioVelardoTheSoundofAI) whose educational video series on deep learning for audio provided the foundational reference and implementation guidance for the autoencoder architectures presented in this work.

The authors also acknowledge the use of Google Gemini and Anthropic Claude as AI assistants during the development and writing of this project. These tools were used for coding guidance, debugging, and writing assistance. All technical implementations, experimental results, analysis, and conclusions are the authors' own work.